YUKUN ZHOU

◇ Hong Kong ◇ yukunzhou450@gmail.com

EDUCATION

City University of Hong Kong (CityU of HK)
Bachelor of Computer Science, Minor in Creative Media, 2019 - 2023

- CGPA: 3.8/ 4.30 (Top 14 out of 144 CS students), average 3.95 for all major courses
- First Class Honor

University of Southern California (USC)
Master of Science, Computer Science - Game Development 2023 - 2025

- CGPA: 3.87/ 4.0

RESEARCH OUTPUT

Manuscript under Review
Anonymous author, Yukun Zhou, Other Authors. About the optimization of Vertex Block Descent Manuscript submitted, 2026.

RESEARCH EXPERIENCE

Research Assistant
Advised by: Dr Chenfanfu Jiang June 2026- Current
Univeristy of California, Los Angeles *Los Angeles*

- Working on Cable Manipulation by developing real-time anisotropic and elasto-plastic rod simulation with non-penetrating contact.
- Deploying on a single-arm Franka robot.

Joint RA–PhD Program - Physics-Based Simulation
Advised by: Dr Zhongkai Zhang, Dr Hongbin Liu July 2025- June 2026
Chinese Academy of Sciences (Hong Kong Institute of Science and Innovation) and Chinese University of Hong Kong *Hong Kong*

- **Manuscript in Preparation**
  Yukun Zhou, Other Authors. Thickness-Aware Rod Insertion Simulation with Rotational Friction. Manuscript in preparation, 2026.
- Finished main experiments and producing a paper draft, about accurate rod-insertion simulation algorithms based on Projective Dynamics, Cosserat rods using Lagrangian constraint-based methods, and modelling thickness-aware rod contact pressure and rotational frictions.
- Worked on the optimization of the Vertex Block Descent project and paper.
- Developed and reproduced sdf softbody contact, DiSect differentiable cuttings, FEM, Projective Dynamics cutting with Low-rank update methods, lung puncturing by Warp.
- Completed human brain 3D reconstruction from MRI images using neural networks.

Summer Research Intern - Fluid Simulation about Neural and Particle Flow Map
Advised by: Dr Bo Zhu June 2024-Sep 2024
Georgia Institute of Technology *Atlanta, USA*

- Replicated Neural Flow Map algorithm by Taichi and develop the real-time 3D stable fluid simulation by OpenGL Compute Shader.
- Developed the algorithm acceleration with Long-range-mapping-short-range-projection approaches on Particle Flow Map method with 80% speed improvement.

Part-time Research Assistant - HCI Research on the Game and Players
Advised by: Dr Zhicong Lu — June 2022-Sep 2023
City University of Hong Kong — *Hong Kong*

- Produced a paper submitted to the top-tier conference as the only first author.
- Completed the research project that contributes to games and ICT4D in HCI and provides insights to enhance their well-being by suggesting design implications and further application scenarios.

WORK EXPERIENCE

Grader of Physical-based Simulation and Animation Course at USC Work as the grader for the course by Dr. Jernej Barbic — Jan 2025-May 2025

Data Science Intern (Full-time, Four Days A Week) Siemens - Mobility, Hong Kong — June 2021-March 2022

- Designed and developed map matching, route matching and traffic flow prediction algorithms and machine learning time series models (bi-LSTM, Encoder-decoder LSTM, ARIMA).
- Database manipulate and data process by Python to support database building and maintaining.

RELATED PROJECTS

Physics-based Simulations Implemented Mass Spring system, Keyframe insertion with Quaternion and Euler approach, Inverse Kinematic with Pseudo Inverse method and Tikhonov regularization method in C++.

Game Development: Lucidmare, PL-23 and Donuts! by Unity Developed Unity 3D programs in C, lead and design in three games: Lucidmare(Demo), Omnis(Demo), Donuts! (Published on Steam)

3D Human Rendering A 3D graphic course group project, achieving URP, hair anisotropy, and skin separable subsurface scattering under Unity engine. Learn and deploy skin separable subsurface scattering in glsl.

Hand-scratching Gesture Recognition Android Application for AR Glasses Advised by: Dr Kening Zhu, *City University of Hong Kong*. Developed an Android application with CNN models to analyze the sound of scratching the AR glasses leg in different ways in real-time to take the recording and real-time recognition function, and control the AR glasses.

NLP for Finance Advised by: Dr Linqi Song, *City University of Hong Kong*. Applied deep learning knowledge and developed the BERT model to analyze the sentiment of large amounts of comments in PyTorch.

AWARDS & HONORS

- Entrance Full Tuition Scholarships — 2019-2023
- Dean's list awards for outstanding academic performance — 2019-2023

SKILLS

Other Technical Skills	AI programming, Drone Programming, Digital Art and Installation, Game Design and Development
Programming Language	Python, C, Java, C++, Processing, JavaScript, HTML, CSS, SQL
Language	Native Mandarin, Fluent English, Basic Cantonese